I/Q imbalance pre-compensation in OFDM systems

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***Abstract*—Numerical results show that the proposed I/Q imbalance pre-compensation schemes improve the bit error probability in an OFDM system.**

***Index Terms*—I/Q imbalance, Pre-distortion, deep learning**

I. Introduction

Recently, New Radio (NR) has emerged as one of the most prominent wireless technologies, designed to meet the increasing demands for higher data rates, lower latency, and massive connectivity. As wireless communication technology is standardized for 5G systems, NR employs advanced techniques to improve spectrum efficiency and service reliability [1]. Among its key features, NR adopts Orthogonal Frequency Division Multiplexing (OFDM) as the underlying modulation scheme. OFDM is a well-known multi-carrier technique that enables high-speed data transmission through parallel data streams. Then, OFDM has been widely employed in various wireless communication standards such as Wireless Local Area Networks (WLANs) [2]-[3].

OFDM offers several advantages, including high spectral efficiency and strong robustness against multi-path fading and Doppler effects. It is also resilient to narrow-band interference and enables high-speed signal processing through Fast Fourier Transform (FFT) techniques, facilitating efficient hardware implementation. Moreover, OFDM systems do not require complex equalizers and shows strong resistance to impulse noise. Due to these characteristics, OFDM remains a widely adopted modulation scheme for ultra-high speed wired and wireless communication systems worldwide, with extensive ongoing standardization efforts based on OFDM technologies. However, OFDM systems are highly sensitive to impairments such as frequency offset, high Peak-to-Average Power Ratio (PAPR), DC offset, in-phase/quadrature imbalance, and phase noise [4]-[5].

Because modern wireless systems such as LTE, 5G, and WiFi rely heavily on OFDM, there is a strong demand for highly integrated, low-cost, and power-efficient OFDM receivers suitable for large-scale deployment. In response to this demand, significant researches have focused on developing compact and cost-effective receiver architectures. Among the developed receiver architectures, zero Intermediate Frequency (zero-IF) and direct-conversion receivers have been widely adopted due to high level of integration and removal of expensive Intermediate Frequency (IF) filters [6]. Compared with the conventional super-heterodyne architecture, these receivers achieve higher integration while avoiding the need for expensive intermediate-frequency (IF) filters. However, while symbol mapping and baseband processing are carried out digitally, the up/down-conversion are implemented in the analog domain, where I/Q imbalance is inevitable. I/Q imbalance arises from the non-ideality in the RF front-end, where the gain balance and orthogonality between the in-phase and quadrature branches are not perfectly maintained. Furthermore, OFDM is highly sensitive to non-linearity in the receiver front-end, due to its high PAPR. Such non-linearity distorts the orthogonality among the subcarriers and results in inter-carrier interference. Therefore, I/Q imbalance mitigation algorithm should be investigated.

There have been several researches to compensate I/Q imbalance [7]-[9]. These studies require additional pilot or reference signals to estimate and compensate the I/Q imbalance, which reduces resource efficiency. Furthermore, most of these techniques focused on post-compensation at the receiver side rather than pre-compensation at the transmitter. Since post-compensation generally results in larger residual errors compared to pre-compensation, further research on pre-compensation techniques is required. In this paper, we propose two pre-distortion schemes to mitigate I/Q imbalance at the baseband. The first scheme is based on the RF signals, which does not require additional pilot and reference signals. The second scheme is based on one of the deep learning algorithms, specifically using a Bi-directional Long Short-Term Memory (BLSTM) [10]-[12]. In this paper, the Bit Error Rate (BER) performance of the proposed schemes is evaluated by simulation. The simulation results demonstrate that the proposed pre-distortion schemes improve the BER performance of the OFDM system under I/Q imbalance conditions.

II. System model

In this section, a system model is described for an OFDM communication system, which is one of the systems with multi-carrier transmission. In the OFDM communication system, a data stream with a high transmission rate is divided into several data streams with a low transmission rate. These data streams are each mapped to a subcarrier, and transmitter can transmit them simultaneously using multiple subcarriers. If N subcarriers are used, the OFDM system can increase the symbol interval by a factor of N. Then, it can be reduced to the relative multi-path spread with respect to the symbol interval by a factor of N.

Fig. 1 shows the block diagram of transmitter in the OFDM communication system. In Fig. 1, input data $d(t)$ is modulated into complex data symbols such as Phase Shift Keying (PSK) or Quadrature Amplitude Modulation (QAM) through a mapper process. After the mapper process, the modulated data

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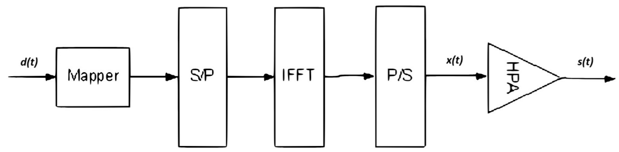

Figure 1. Block diagram of transmitter in OFDM communication system

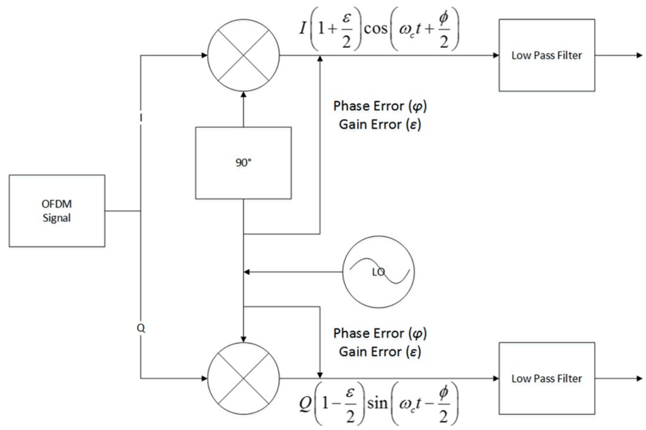

Figure 2. I/Q imbalance of the local oscillator

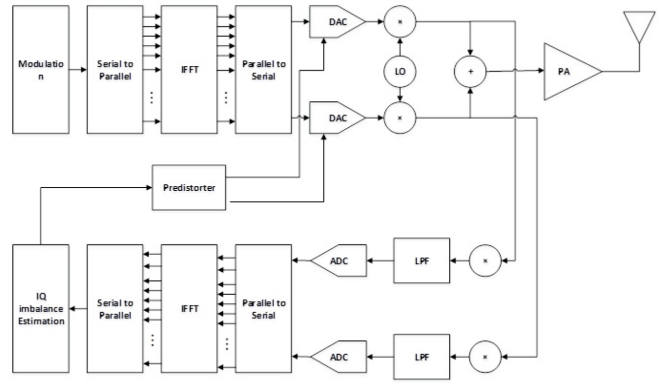

Figure 3. Transmitter structure with I/Q imbalance pre-distorter based on RF signal.

symbols are mapped to multiple subcarriers through a Serial to Parallel (S/P) process, and modulated into OFDM symbols by Inverse Fast Fourier Transform (IFFT). The time-domain OFDM symbols $x(t)$ can be expressed as

$$x(t) = \frac{1}{N}\sum_{n=0}^{N-1} X_n e^{j2\pi\Delta f t}, for\ 0 \le t < NT, \tag{1}$$

where N and X_n is the number of subcarriers and mapped data symbol for n-th subcarrier, respectively. In (1), T is data duration and Δf is subcarrier spacing, which is expressed as $1/NT$. In an OFDM system, Inter Symbol Interference (ISI) and Inter Carrier Interference (ICI) can exist by fading channels. To reduce the performance degradation of ISI and ICI, the Cyclic Prefix (CP) is inserted between OFDM symbols. After the CP insertion, the time-domain OFDM symbols are up-converted to RF signals $s(t)$ onto the Local Oscillator (LO) of High Power Amplifier (HPA) and transmitted.

In (1), the time-domain OFDM signals $x(t)$ can be expressed as a sum of in-phase signal $I(t)$ and quadrature signal $Q(t)$ ($x(t) = I(t) + jQ(t)$). In the up-conversion process, in-phase and quadrature signals are mapped onto cosine and sine waves of Local Oscillator (LO), respectively. Then, the RF signal $s(t)$ can be expressed as

$$s(t) = I(t) \cdot cos(w_c t) + Q(t) \cdot sin(w_c t), \tag{2}$$

where w_c is the carrier frequency of the RF signal.

However, in a practical system, the cosine and sine waves of LO do not have the same gain and phases, which is called as I/Q imbalance. Fig. 2 shows the I/Q imbalance of the LO. The signal with I/Q imbalance $\hat{s}(t)$ can be expressed as

$$\begin{aligned}\hat{s}(t) = {} & I(t)\left(1+\frac{\varepsilon}{2}\right) cos\left(w_c t + \frac{\varphi}{2}\right) \\ & + Q(t)\left(1-\frac{\varepsilon}{2}\right) sin\left(w_c t - \frac{\varphi}{2}\right),\end{aligned} \tag{3}$$

where ε and φ are the gain and phase errors, respectively.

III. I/Q IMBALANCE PRE-DISTORTION SCHEMES

The gain and phase errors from the I/Q imbalance of LO degrade the performance of system. To reduce the degradation, pre-distortion can be applied in a transmitter. In this section, two I/Q imbalance pre-distortion schemes are presented.

A. RF signal based pre-distortion of I/Q imbalance

Fig. 3 shows the transmitter structure with I/Q imbalance pre-distorter based on the RF signal. For the pre-distortion of I/Q imbalance, the feedback process is performed with the RF signal as input. A transmitter down-converts the RF signal using the LO used in the up-conversion process. The down-conversion signal $\tilde{s}(t)$ can be expressed as the sum of in-phase signal $\tilde{I}(t)$ and quadrature $\tilde{Q}(t)$ ($\tilde{x}(t) = \tilde{I}(t) + j\tilde{Q}(t)$). Then, $\tilde{I}(t)$ and $\tilde{Q}(t)$ can be expressed as

$$\begin{aligned}\tilde{I}(t) = {} & \int_0^t \hat{s}(\tau)\left(1+\frac{\varepsilon}{2}\right) cos\left(w_c\tau + \frac{\varphi}{2}\right) + n(\tau) d\tau \\ = {} & \int_0^t I(\tau)\left(1+\frac{\varepsilon}{2}\right)^2 cos^2\left(w_c\tau + \frac{\varphi}{2}\right) d\tau \\ + {} & \int_0^t Q(\tau)\left(1-\frac{\varepsilon^2}{4}\right) cos\left(w_c\tau + \frac{\varphi}{2}\right) sin\left(w_c\tau + \frac{\varphi}{2}\right) d\tau \\ + {} & \int_0^t n(\tau) d\tau \\ = {} & I_L(t)\left(1+\frac{\varepsilon}{2}\right)^2 - Q_L(t)\left(1-\frac{\varepsilon^2}{4}\right) sin\varphi + n_L(t),\end{aligned} \tag{4}$$

$$
\begin{aligned}
\tilde{Q}(t) &= \int \hat{s}(t)\left(1-\frac{\varepsilon}{2}\right) sin\left(w_c t-\frac{\varphi}{2}\right)+n(t)dt \\
&= \int I(t)\left(1-\frac{\varepsilon^2}{4}\right) cos\left(w_c t+\frac{\varphi}{2}\right) sin\left(w_c t-\frac{\varphi}{2}\right) dt \\
&+ \int Q(t)\left(1-\frac{\varepsilon}{2}\right)^2 sin^2\left(w_c t-\frac{\varphi}{2}\right) dt \\
&+ \int n(t)dt \\
&= Q_L(t)\left(1-\frac{\varepsilon}{2}\right)^2 - I_L(t)\left(1-\frac{\varepsilon^2}{4}\right) sin\varphi + n_L(t).
\end{aligned} \tag{5}
$$

where $I_L(t)$, $Q_L(t)$ and $n_L(t)$ represent the I and Q signals and internal thermal noise after Low Pass filtering, respectively.

In (4) and (5), ε^2 can be negligible for $0 < \varepsilon < 0.1$. Then, from (4) and (5), the gain error ε_T and the phase error φ_T can be estimated as

$$\varepsilon_T = \frac{I(t)\tilde{I}(t) - Q(t)\tilde{Q}(t) - (I^2(t) - Q^2(t))}{I^2(t) + Q^2(t)}, \tag{6}$$

$$\varphi_T = sin^{-1}\left(\frac{2I(t)Q(t) - (I(t)\tilde{Q}(t) - Q(t)\tilde{I}(t))}{I^2(t) + Q^2(t)}\right). \tag{7}$$

If the estimated gain error ε_T and phase error φ_T are applied in the up-conversion process, the effect of I/Q imbalance can be reduced.

B. Deep learning based pre-distortion of I/Q imbalance

In the previous subsection, an I/Q imbalance pre-distortion scheme based on RF signals was proposed. Under ideal conditions, this scheme shows the BER performance similar to the case of no I/Q imbalance. However, the performance degradation of pre-distortion can be observed in environments with noise in the internal circuitry. This performance degradation is considered to result from the process of converting the RF signal through the DAC and the subsequent internal feedback loop involved in the RF signal-based pre-distortion scheme. Then, in this subsection, the deep learning based pre-distortion of I/Q imbalance is proposed, which does not involve the internal feedback or DAC processes.

Deep learning algorithms such as Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTMs) are well-known to provide highly accurate modeling of data and systems in various communication systems. In this paper, we introduce BLSTM, which is one of the deep learning algorithms. Based on BLSTM, we propose a novel I/Q imbalance pre-distortion technique.

Unlike traditional LSTM, BLSTM processes the information in both the forward and backward directions (Bi-directions), considering both past and future patterns. For example, when processing the data x_t at the current time t, the model can simultaneously analyze the influence of past data $(x_{t-1}, x_{t-2}, ...)$ and future data $(x_{t+1}, x_{t+2}, ...)$ on the current data x_t [10]-[12].

The BLSTM architecture for I/Q imbalance compensation is designed as follows. At the input layer, the real and imaginary parts of the complex signal affected by I/Q imbalance are separately fed into the network. The temporal dependency is learned by computing hidden states in both forward and backward directions of BLSTM. The forward and backward hidden states, denoted by $\overrightarrow{h}_t$ and $\overleftarrow{h}_t$, are expressed as

$$\overrightarrow{h}_t = LSTM_{forward}(x_t, \overrightarrow{h}_{t-1}), \tag{8}$$

$$\overleftarrow{h}_t = LSTM_{backward}(x_t, \overleftarrow{h}_{t+1}). \tag{9}$$

Then, the combined hidden state h_t, obtained by combining the forward and backward hidden states, can be expressed as

$$h_t = [\overrightarrow{h}_t, \overleftarrow{h}_t]. \tag{10}$$

The training of the BLSTM model is performed as follows: First, the pairs of signals, each consisting of a signal with I/Q imbalance and its corresponding ideal target signal, are prepared to form the training dataset. Second, the loss function L is defined as the Mean Squared Error (MSE) between the ideal target signal and the compensated signal. Then, the loss function can be expressed as

$$L = \frac{1}{N}\sum_{i=1}^{N}(|I_{i,ideal} - I_{i,comp}|^2 + |Q_{i,ideal} - Q_{i,comp}|^2), \tag{11}$$

where N is the number of subcarriers. $I_{i,org}$ and $Q_{i,org}$ are in-phase and quadrature part of ideal target signal, respectively. On the other hand, $I_{i,comp}$ and $Q_{i,comp}$ are in-phase and quadrature part of the compensated signal, respectively. Finally, the model is trained using the training dataset and the defined loss function. Through the optimization process, the model learns to effectively compensate I/Q imbalance by minimizing the error between the ideal and the compensated signals.

For I/Q imbalance pre-distortion, the use of BLSTM algorithm offers several advantages. It can effectively compensate non-linear I/Q imbalance, which is challenging for conventional linear compensation algorithms. Moreover, BLSTM is capable of capturing the time-dependent changes in I/Q imbalance, making it suitable for dynamically changing environments. However, the BLSTM-based I/Q imbalance pre-distortion also presents certain limitations. It requires a large amount of training data to achieve reliable performance, and the training process can be computationally intensive and time-consuming. In addition, the implementation may demand more computational resources compared to traditional compensation techniques.

Fig. 4 shows the stacked BLSTM architecture for creating an I/Q imbalance pre-distorter. As shown in Fig. 4, the proposed BLSTM model is composed of two parallel branches, which process the in-phase and quadrature components separately. It processes the input sequence bi-directionally, considering past, present, and future information simultaneously. This approach enables the model to learn time-series data more effectively. Each branch is composed of two BLSTM

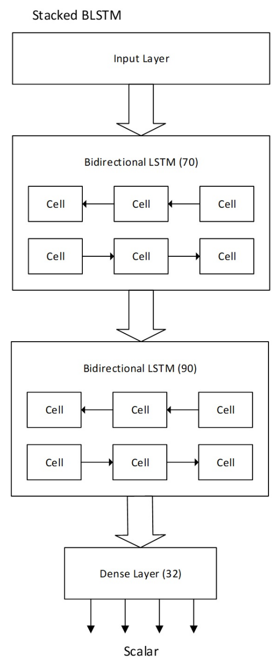

Figure 4. Initial design of stacked BLSTM for creating an I/Q imbalance pre-distorter

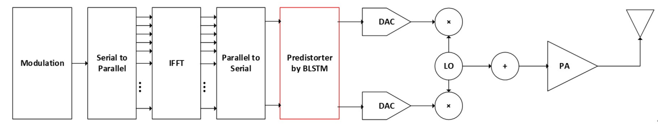

Figure 5. The example of transmitter structure with I/Q imbalance pre-distorter by BLSTM

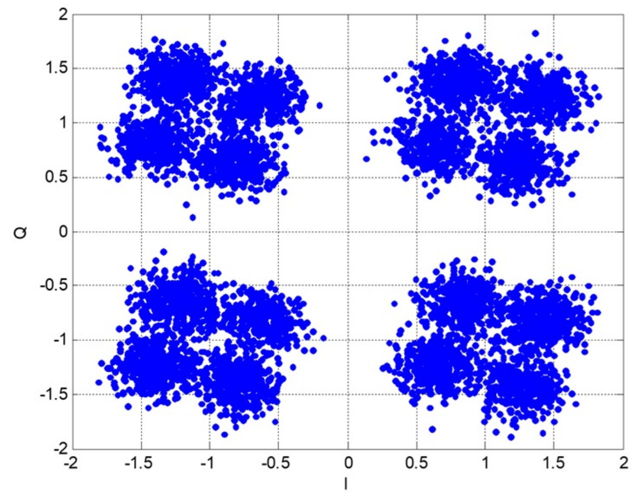

Figure 6. Signal constellation in an AWGN channel with both amplitude and phase imbalances

layers followed by a dense layer. The dense layer following the BLSTM applies linear transformation and non-linear activation, calculating each output neuron as the sum of weighted inputs. In this paper, considering the system complexity and hardware limitations, the OFDM system is designed with an FFT size of 32.

The proposed BLSTM model is composed of input layer, two BLSTM hidden layers, and output layer. The input layer serves as the entry point for data into the network, defining and receiving the shape of the input data before the model begins learning and prediction. The first and second BLSTM hidden layers contain 70 and 90 units, respectively. The output layer predicts 32 independent continuous values, with each represented by a single real-valued neuron output, which are treated as uncorrelated scalar predictions.

To train the BLSTM model, the Root Mean Square Propagation (RSMProp) optimizer was used. The batch size was set to 32, and training was conducted over 60 epochs. The training dataset consisted of a total 3,200,00 OFDM symbols, which were split into training (2,560,000 symbols) and test (640,000 symbols) sets at a ratio of 4:1. The performance of the training model was evaluated using the MSE. On the test set, the model achieved an MSE of 1.2×10^{-4}. In particular, the model demonstrated stable pre-distortion performance even for I/Q imbalance parameters that were not included in the training data, validating its strong generalization capability. These experimental results confirm that the proposed BLSTM architecture is effective for I/Q imbalance pre-distortion and achieves the performance suitable for practical system implementation.

Fig. 5 shows the example of OFDM transmitter structure including the BLSTM used for I/Q imbalance pre-distortion. In Fig. 5, the BLSTM is applied after the Parallel-to-Serial (P/S) conversion. During the BLSTM processing, the OFDM symbols are pre-distorted to mitigate the I/Q imbalance and then up-converted to the RF signal.

IV. Simulation Results

In this section, simulation results are presented. Table I shows the parameters used in a simulation. The OFDM signal is modulated by Quadrature Phase Shift Keying (QPSK). The number of subcarriers is set to 32 and the AWGN and Rician fading channels are considered. For BLSTM training, the input and target data are set to signal with I/Q imbalance and the corresponding ideal signals, respectively. The BLSTM training is performed independently for the I and Q components. The lengths of the first and second layers of BLSTM are set to 50 and 70, respectively.

A. *Signal constellation under I/Q imbalance*

Figs. 6 and 7 show the signal constellations in an AWGN channel with I/Q imbalance and after compensation of the I/Q imbalance using BLSTM based pre-distortion, respectively. In Figs. 6-7, the E_b/N_0 is set to 15 dB. The constellation points become dispersed when either amplitude or phase imbalance is present individually. However, the constellation is both dispersed and rotated when both imbalances are present, as shown in Fig. 6. Compared to other RF impairments such as phase noise and non-linear amplification, I/Q imbalance is typically modeled with discrete amplitude and phase error parameters. This imbalance not only causes rotation of the constellation but also leads to the formation of multiple district

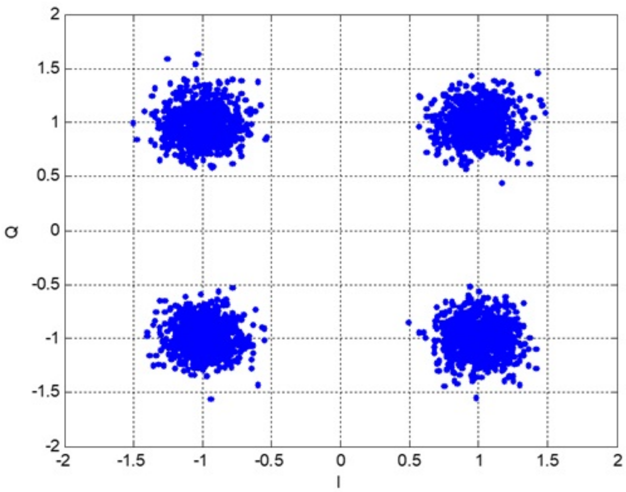

Figure 7. Signal constellation in an AWGN channel after I/Q imbalance compensation

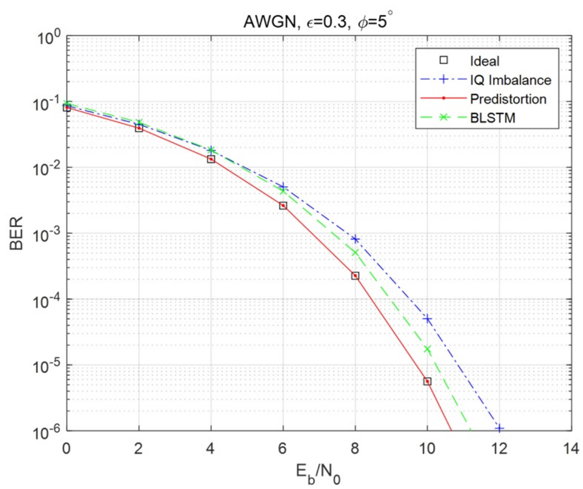

Figure 8. BER performance with and without I/Q imbalance pre-distortion in an AWGN channel

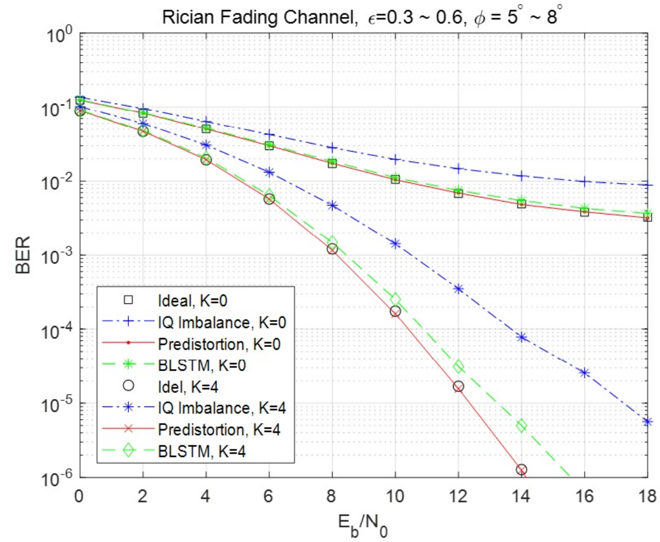

Figure 9. BER performance with and without I/Q imbalance pre-distortion in Rician fading channel

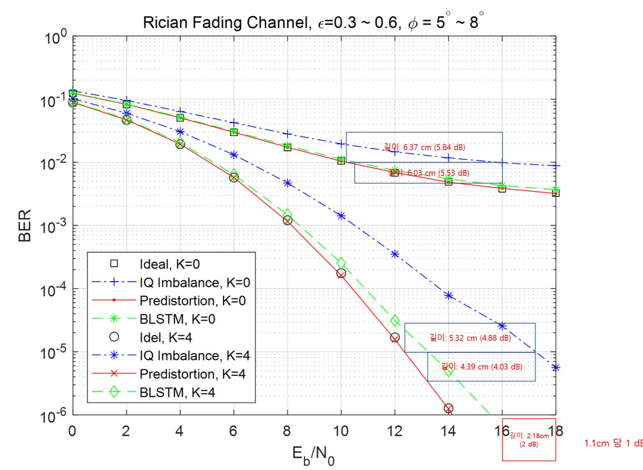

Figure 10. Performance gain

clusters, rather than a simple spreading of points. From the results in Figs. 6 and 7, it can be observed that the constellation cloud is significantly reduced after the compensation of I/Q imbalance using BLSTM based pre-distortion.

B. BER performance

Fig. 8 shows the BER performance in an AWGN channel for two proposed pre-distortion schemes. For comparison, the results for the uncompensated case and the ideal case (no I/Q imbalance) are also plotted. The BLSTM based pre-distortion model was trained on signal with I/Q imbalance parameters of $\varepsilon = 0.3$ and $\varphi = 5\,°$. For the BER of 10^{-5}, the RF signal based and BLSTM based pre-distortion schemes achieve approximately 1.2 dB and 0.8 dB performance improvements compared to the uncompensated case, respectively.

Fig. 9 shows the BER performance of two proposed pre-distortion schemes in an Rician channel with K factors of 0 and 4. For comparison, the results for the uncompensated case and the ideal case (no I/Q imbalance) are also plotted. The BLSTM based pre-distortion model was trained on signal with I/Q imbalance parameters of $\varepsilon = 0.3 \sim 0.6$ and $\varphi = 5 \sim 8\,°$. For the BER of 10^{-2}, the RF signal based and BLSTM based pre-distortion schemes achieve approximately 6.0 dB performance improvements, compared to the uncompensated case under a Rician fading channel with a K factor of 0. For the BER of 10^{-5}, the RF signal based and BLSTM based pre-distortion schemes achieve approximately 4.5 dB and 4.0 dB performance improvements, respectively, compared to the uncompensated case under a Rician fading channel with a K factor of 4.

From the results of Figs. 8 and 9, the RF signal based pre-distortion shows the results close to those obtained under ideal I/Q balance case. In addition, about 0.5 dB performance degradation is observed in the BLSTM-based pre-distortion scheme compared to the RF signal based scheme. However, the performance of the BLSTM-based pre-distortion scheme could be further improved by using more refined training data and the application of optimal layers and parameters.

V. Conclusion

Performance

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